SOFTWARE ENGINEERING LAB — LAB 7

Fuzzing for Input Validation Bugs

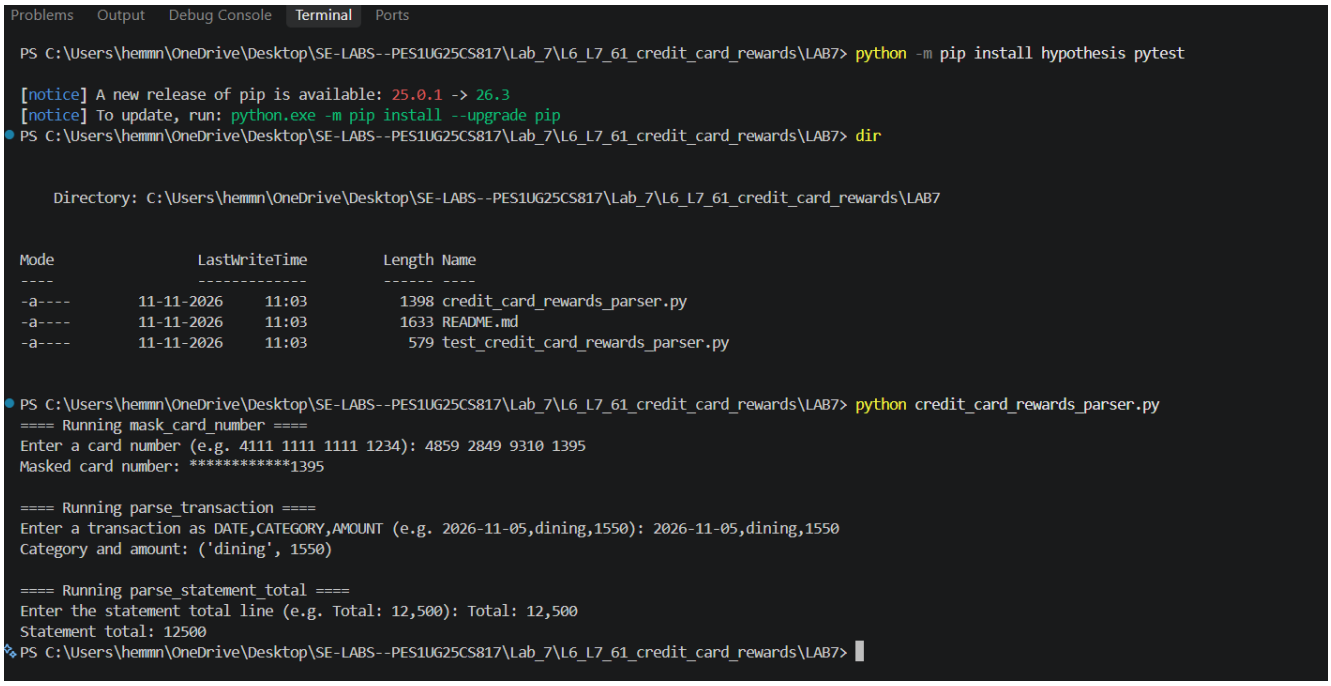
Student Name: Hemanth V
SRN: PES1UG25CS817
Project: Credit Card Rewards Parser

This report records the original program behavior, bugs detected by Hypothesis-based tests, and the successful run after input validation was added.

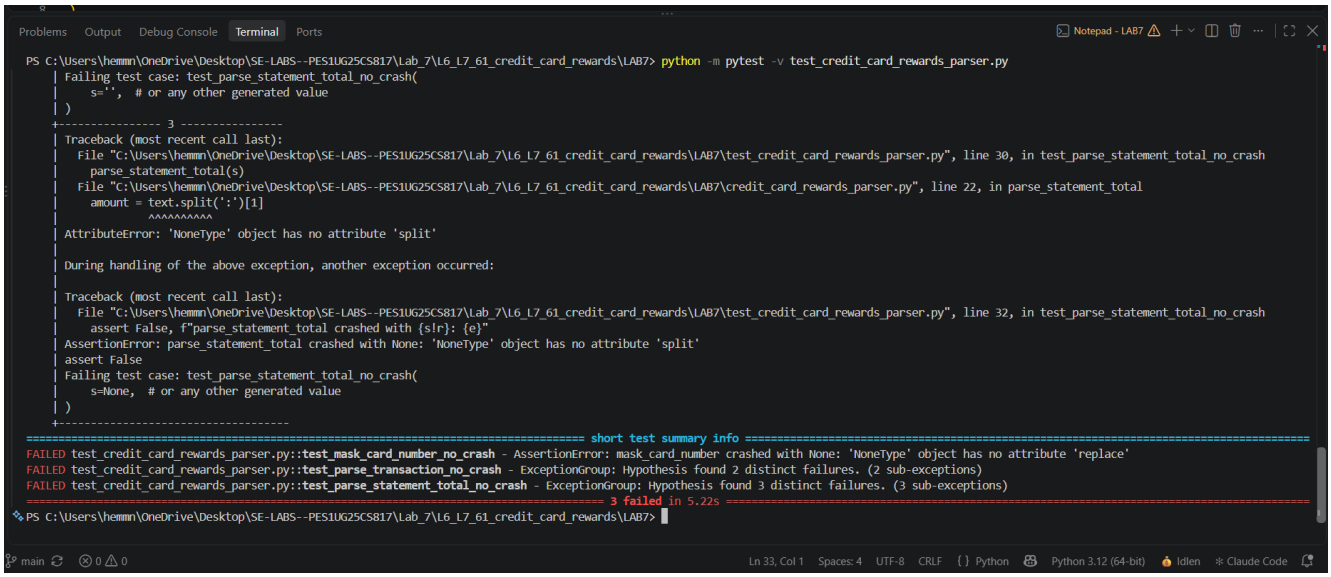
1. Test Evidence

The following screenshots document the progression from the original implementation to the fixed implementation.

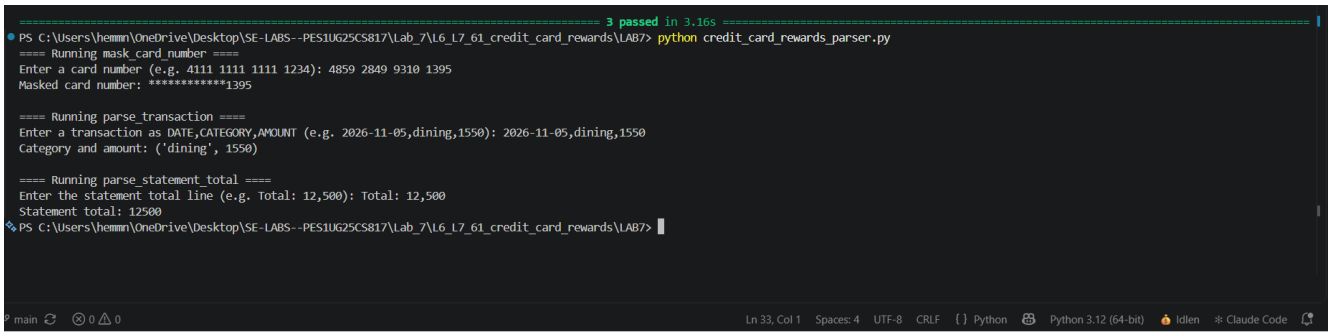
SS1 — Original application run with valid sample inputs



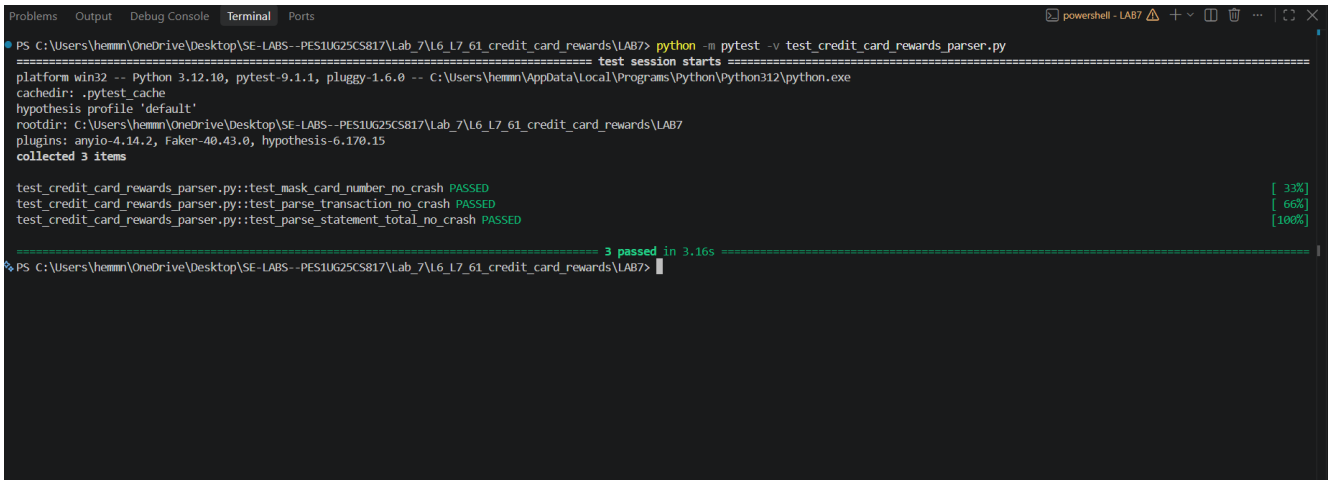
SS2 — Hypothesis tests exposing bugs in the original parser



SS3 — Fixed application run with valid sample inputs



SS4 — Hypothesis tests passing on the fixed parser



2. Implementation Summary

Functions tested

• **mask_card_number(number):** masks card-number characters while handling non-string and empty inputs safely.
• **parse_transaction(line):** validates the input type and expected comma-separated fields; invalid data returns a safe fallback tuple.
• **parse_statement_total(text):** checks for a string and a colon-delimited amount, then safely converts the amount to an integer.

Testing approach

Hypothesis generates many text values, including None, to search for inputs that cause exceptions. The original parser failed on generated invalid inputs. After defensive checks and safe conversion were added, the three no-crash tests passed.

3. Reflection

1. What bugs did fuzz testing reveal?
Hypothesis testing revealed that the original parser could crash when given None or malformed strings. For example, mask_card_number() failed when calling replace() on None, while parse_statement_total() failed when calling split() on None.

2. How did you fix the bugs?
I added input-type checks, validation of the expected input format, and safe integer conversion. Invalid inputs now return fallback values instead of causing exceptions.

3. What did you learn from fuzz testing?
I learned that testing only valid inputs is insufficient. Hypothesis automatically generates many inputs, including unexpected values, helping identify edge cases that manual testing might miss.

Note: These reflection prompts are suggested because the supplied handout screenshot refers to three questions but does not display their exact wording.